

Yongjin Choi, Ph.D.

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EDUCATION

Ph.D. in Civil Engineering <i>University of Texas at Austin, Department of Civil, Architectural and Environmental Engineering, USA</i> • Specialty area: geotechnical engineering • Dissertation: Forward and Inverse Modeling of Granular Flows Using Differentiable Graph Neural Network Simulator • Advisor: Prof. Krishna Kumar	2024
M.S. in Civil Engineering <i>Pusan National University, Department of Civil and Environmental Engineering, South Korea</i> • Specialty area: geotechnical engineering • Thesis: Stiffness Evaluation of Open-Graded Aggregate Base Corresponding to Compaction Level Based on Plate Load Tests • Advisor: Prof. Jaehun Ahn	2019
B.S. in Civil Engineering <i>Pusan National University, Department of Civil and Environmental Engineering, South Korea</i>	2017

APPOINTMENTS

Assistant Professor (Incoming) <i>University of Massachusetts Amherst, USA</i>	09/2026
Postdoctoral Fellow (Supervisor: Seunghwa Ryu) <i>Korea Advanced Institute of Science & Technology (KAIST), South Korea</i>	2025 – Current
Affiliated Research Fellow (Supervisor: Jorge Macedo) <i>Georgia Institute of Technology, USA</i>	2024 – Current
Postdoctoral Fellow (Supervisor: Jorge Macedo) <i>Georgia Institute of Technology, USA</i>	2024 – 2025
Graduate Research Assistant (Supervisor: Krishna Kumar) <i>University of Texas at Austin, USA</i>	2021 – 2024
Graduate Research Assistant (Supervisor: Jaehun Ahn) <i>Pusan National University, South Korea</i>	2017 – 2020

PEER-REVIEWED PAPERS (* Corresponding author)

1. (Under review) Ryou, J. E., Gang, S., **Choi, Y.**, Lee, E. H. “Microstructure-Informed Prediction of Hydraulic Conductivity Using Pore Network Modeling and Interpretable Surrogate Learning”, *Journal of Hydrology*
2. (Under review) **Choi, Y.**, Macedo, J.* (2025) “Differentiable graph neural network simulator for the back-analysis of post-liquefaction residual strength from flow failure runout.” *Canadian Geotechnical Journal* [\[Pre-print\]](#)
3. **Choi, Y.**, Macedo, J.*, Liu, C. (2025) “Differentiable graph neural network simulator for forward and inverse modeling of multi-layered slope system with multiple material properties.” *Soils and Foundations*, 66(3), 101773 [\[Link\]](#)
4. **Choi, Y.**, Lee, S.* (2026) “Accelerating stochastic simulation of post-failure landslide runout using a random graph neural network-based simulator” *Computers & Geotechnics*, 195, 108104 [\[Link\]](#)
5. **Choi, Y.**, Liu, C., Macedo, J.* (2025) “Operator learning for consolidation: An architectural comparison for DeepONet variants.” *Computers and Geotechnics*, 194, 108017 [\[Link\]](#)
6. Ryou, J. E., Gang, S., **Choi, Y.**, Dai, S., Jung, J.* (2026) “The Interplay Between Pore and Particle Characteristics in Sands Based on Computed Tomographic Image Analyses.” *Powder Technology*, 122150 [\[Link\]](#)
7. Bae, B., **Choi, Y.***, Jung, H., Ahn, J.* (2025) “Tunnel lining segmentation from ground-penetrating radar images using advanced single-and two-stage object detection and segmentation models.” *Computer-Aided Civil and Infrastructure Engineering*, 40(22), 3580-3592 [\[Link\]](#)

8. Kim, T., Shahriar, A. R., Lee, W.-D.*, **Choi, Y.**, Kwon, S., & Gabr, M. A. (2025) "Field data-based prediction of local scour depth around bridge piers using interpretable machine learning." *Transportation Geotechnics*, 52, 101567. [\[Link\]](#)
9. **Choi, Y.**, Bae, B., Han, T. H., Ahn, J.* (2024) "Application of Mask R-CNN and YOLOv8 Algorithms for Concrete Crack Detection." *IEEE Access*. [\[Link\]](#)
10. **Choi, Y.***, Kumar, K. (2024) "Inverse Analysis of Granular Flows Using Graph Neural Network-Based Simulator." *Computers and Geotechnics*, 171, 106374. [\[Link\]](#) [\[Code\]](#)
11. **Choi, Y.**, Nguyen, H. T., Han, T. H., Choi, Y., Ahn, J.* (2024) "Sequence Deep Learning for Seismic Ground Response Modeling: 1D-CNN, LSTM, and Transformer Approach." *Applied Science*, 14(15), 6658 [\[Link\]](#) [\[Code\]](#)
12. Lee, Y., Lee, D., Lee, H., Choe, H. S., Kim, J. H., **Choi, Y.***, Ahn, J.* (2024). "Removal Efficiency of Bottom Ash and Sand Mixtures as Filter Layers for Fine Particulate Matter." *Materials*, 17(11), 2749. [\[Link\]](#)
13. **Choi, Y.***, Kumar, K. (2024). "Graph Neural Network-based Surrogate Model for Granular Flows." *Computers and Geotechnics*, 116, 106015. [\[Link\]](#) [\[Code\]](#)
14. Kumar, K., **Choi, Y.** (2023). "Accelerating Particle and Fluid Simulations with Differentiable Graph Networks for Solving Forward and Inverse Problems." *The International Conference for High Performance Computing, Networking, Storage and Analysis (Top-tier conference in computer science)*, Denver, USA. [\[Link\]](#)
15. Lee, Y., **Choi, Y.**, Ahn, D., Ahn, J*. (2021). "Prediction Models Based on Regression and Artificial Neural Network for Moduli of Layers Constituted by Open-Graded Aggregates." *Materials*, 14(5), 1199. [\[Link\]](#)
16. **Choi, Y.**, Ahn, D., Lee, Y., Ahn, J*. (2020). "Compaction Quality Monitoring of Open-Graded Aggregates by Light Weight Deflectometer (LWD) and Soil Stiffness Gauge (SSG)." *Sustainability*, 12(6), 2521. [\[Link\]](#)
17. **Choi, Y.**, Ahn, D., Nguyen, T. H., Ahn, J*. (2018). "Assessment of Field Compaction of Aggregate Base Materials for Permeable Pavements Based on Plate Load Tests." *Sustainability*, 10(10), 3817. [\[Link\]](#)

PEER-REVIEWED JOURNAL PAPERS (Korean Journals) (* Corresponding author)

1. **Choi, Y.**, Son, S. (2024). "Sensitivity analysis of factors influencing the settlement of buried pipes installed beneath roads using explainable machine learning techniques." *Journal of the Korean Asphalt Institute*, 14(2), 269-278. [\[Link\]](#)
2. Jong, J., **Choi, Y.**, Ahn, J. (2021). "Evaluation of p -y Curves for Group Piles in Cohesive Soil Based on 3D Numerical Analysis." *Journal of the Korean Society of Hazard Mitigation*, 21(6), 193-200. [\[Link\]](#)
3. **Choi, Y.**, Ahn, J*. (2020). "Evaluation of p -y Curves and p -multiplier of Pile Groups Corresponding to Sand Properties Change Based on 3D Numerical Analysis." *Journal of the Korean Society of Hazard Mitigation*, 20(4), 207-2017. [\[Link\]](#) [\[Code\]](#)
4. **Choi, Y.**, Nguyen, T. H., Ahn, J*. (2019). "Resilient Modulus and Permanent Deformation of 40 mm Open-Graded Aggregates based on Repeated-Load Triaxial Tests." *Journal of the Korean Society of Hazard Mitigation*, 19(7), 367-374. [\[Link\]](#)
5. **Choi, Y.**, Oh, J., Han, S. I., Ahn, J., Shin, H. S*. (2017). "Elastic Modulus and Layer Coefficient of Permeable Block Pavements Based on Plate Load Tests." *Journal of the Korean Geotechnical Society*, 33(12), 75-80. [\[Link\]](#)

CONFERENCE PRESENTATIONS

1. Alam, M. M. (Poster presenter), **Choi, Y.**, Yousaf, A., Seong, W. K., Ryu, S., Ruoff, R. S. (2026). "Geometry of a Non-Centrosymmetric 'Truncated Octahedral' Diamond Particle." *Korean Institute of Metals and Materials Spring Conference 2026*, Jeju, South Korea.
2. **Choi, Y. (Oral presenter)**, Cornejo, R., Macedo, J., Kumar, K. (2024). "AI/MPM-based assessment of Dam break Failure modes and Runout." TAILENG Annual Meeting, Denver, Colorado.
3. **Choi, Y. (Poster presenter)**, Kumar, K. (2024). "Three-dimensional granular flow simulation using graph neural network-based learned simulator." *Geo-Congress 2024*, Vancouver, Canada. [\[Link\]](#)
4. Kumar, K., Mehta, C., **Choi, Y. (Oral presenter)** (2023). "Graph Network Simulator and Differentiable Material Point Method." *Advances in Computational Mechanics 2023*, Austin, USA.
5. **Choi, Y. (Oral presenter)**, Kumar, K. (2023). "Slope Inverse Analysis Using Accurate and Generalizable Data-Driven Surrogate Granular Flow Simulator." *17th National Congress on Computational Mechanics*, Albuquerque, USA.
6. **Choi, Y. (Oral presenter)**, Kumar, K. (2022). "A Machine Learning Approach to Predicting Pore Pressure Response in Liquefiable Sands Under Cyclic Loading." *Geo-Congress 2023*, Los Angeles, USA. [\[Link\]](#) [\[Code\]](#)
7. **Choi, Y. (Poster presenter)**, Lee, S, Jung, J., Oh, J., Ahn, J. (2018). "Coefficient of Subgrade Reaction for the Permeable Block and Base System at Korea GI and LID Center." *Proceedings of the 5th GeoChina International Conference*, Hangzhou, China. [\[Link\]](#)

8. **Choi, Y. (Oral presenter)**, Ahn, D., Nguyen, H. T., Ahn, J. (2018). “Measurement of Elastic Modulus of 40 mm Crushed Stone Aggregate with Field Compaction Effort.” *Korean Society of Railway Spring Conference 2018*, Jeju, South Korea.
9. **Choi, Y. (Oral presenter)**, Jalmasco, M. M., Ahn, J. (2017). “Permeability Degradation of Permeable Pavement Material due to Clogging.” *JpGU-AGU Joint Meeting 2017*, Chiba, Japan.

TEACHING EXPERIENCES

Guest Lecturer **Fall 2025**

Deep Learning and Its Application to Geotechnical Systems, Pusan National University, South Korea

- Delivered a two-hour lecture on physics-informed neural networks and graph neural networks for geotechnical modeling for graduate students.

Assistant Instructor **Spring 2025**

Numerical Modeling and Data Science for Geotechnics (CEE-8813-H), Georgia Institute of Technology, USA

- Designed and delivered lectures and course materials on material point methods [\[link\]](#), finite element methods [\[link\]](#), and regression and classification machine learning models for geotechnical engineers, which are publicly available through open-source websites.

Classroom Project Mentor **Spring 2023**

Scientific Machine Learning (CE 397), University of Texas at Austin, USA

- Mentored a graduate student team in developing a graph neural network-based simulator for slope failure, resulting in an “A” grade for the final project report.

Teaching Assistant **Spring 2023**

Introduction to Computer Methods (CE 311K), University of Texas at Austin, USA

- Hosted weekly laboratory sessions for hands-on Python coding activities for engineering problems, held regular office hours, and managed grading tasks.

Teaching Assistant **Spring 2021**

Probability and Statistics for Civil Engineers (CE 311S), University of Texas at Austin, USA

- Designed and hosted weekly laboratory sessions with hands-on problem-solving activities, held regular office hours, and managed grading tasks.

AWARDS & GRANTS

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| • Community of Practice (CoP) program fellowship, Korea Institute of Human Resources Development in Science and Technology (KIRD) | 2026 |
| • Young Scholar Award, Korean Geotechnical Society | 2023 |
| • Award for 2nd best poster presentation in graduate student symposium of civil, architectural, and environmental engineering, University of Texas at Austin | 2022 |
| • Frontera compute resource award under the supervision of Dr. Kumar, Texas Advanced Computing Center (TACC). | 2021 |
| • Award for outstanding graduate student research paper, Brain Korea 21 Plus | 2019 |
| • Fellowship from “Earthquake Disaster Prevention Human Resource Development”, Korean Ministry of the Interior and Safety | 2017 – 2018 |
| • Scholarship for academic excellence, Korea Student Aid Foundation | 2015 – 2016 |

INVITED TALKS

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| • Invited speaker, “Graph neural network simulator (GNS) for granular flow modeling”, RIT seminar, Department of Civil and Environmental Engineering, Yonsei University, South Korea | 05/2026 |
| • Invited speaker, “Graph neural network simulator (GNS) for granular flow modeling and differentiable constitutive modeling”, Department of Civil, Environmental and Architectural Engineering, Korea University, South Korea | 03/2026 |
| • Invited speaker, “Physics-aware AI for accelerating geohazard modeling and differentiable constitutive modeling”, School of Civil Engineering, Chungbuk National University, South Korea | 01/2026 |
| • Invited speaker, “Physics-aware AI for accelerating geohazard modeling and differentiable constitutive modeling”, Korea Institute of Geoscience and Mineral Resources (KIGAM), South Korea | 12/2025 |

- Invited speaker, “GNN-based landslide post failure analysis and neural operators for soil consolidation modeling”, Department of Mechanical Engineering, Ulsan National Institute of Science & Technology (UNIST), South Korea **12/2025**
- Invited speaker, “GNN-based surrogate modeling”, Department of Mechanical Engineering, Ulsan National Institute of Science & Technology (UNIST), South Korea **11/2025**
- Invited speaker, “Physics-aware AI for geohazard modeling”, GeoKAIST Seminar, Department of Civil and Environmental Engineering, Korea Advanced Institute of Science & Technology (KAIST), South Korea. **10/2025**

OUTREACH & SERVICE

Advisory Board in Revolutionizing Engineering Departments (RED) program, Georgia Institute of Technology **2025**

- Engaged in ongoing educational innovation to incorporate computational thinking and artificial intelligence into the Civil and Environmental Engineering program.

Machine Learning Model Developer in DesignSafe Academy/NHERI Hackathon **2022**

- Participated in team hackathon project for machine learning and high-performance computing held in Texas Advanced Computing Center, and developed p-wave arrival time detector from earthquake source using LSTM and CNN [\[Code\]](#)

President of Korean Civil Engineering Student Association at University of Texas at Austin **2022**

- Led and represented the organization, holding semester-wise social gatherings to foster a supportive environment for Korean civil engineering students.

Journal Reviewer

- Journal of Computing in Civil Engineering | Journal of Traffic and Transportation Engineering | Journal of Rock Mechanics and Geotechnical Engineering | Earthquake Spectra | Chemical Engineering Journal **2021-2026**

SOFTWARE DEVELOPMENT

Learn-MPM-2D **2024**

- Developed an open-source learning package for the 2-D material point method [\[link\]](#), guiding learners from core theory to hands-on practice in a beginner-friendly interactive Python notebook environment.

Cbgeopy: material point method model generator [\[Code\]](#) **2024**

- Developed a material point method model generator that supports various geometries from simple columns to multi-layered 3D topography, which can be integrated with CB-Geo MPM.

GNS: A generalizable Graph Neural Network-based simulator for particulate and fluid modeling [\[Code\]](#) **2023**

- Contributed as a core developer to implement mesh graph network for fluid modeling, multi-material features, and code maintenance work.

PROFESSIONAL LICENSES & CERTIFICATIONS

- Advanced teaching certificate, Center for Teaching and Learning, University of Texas at Austin, USA **2024**
- Engineering in Train (EIT), Civil Engineering, NCEES, USA **2023**
- Machine Learning, Stanford University and Coursera [\[Link\]](#) **2022**
- License of Engineer (FE), Civil Engineering, Human Resources Development Service of Korea **2016**

REFERENCES

Dr. Seunghwa Ryu, Ph.D. (Postdoctoral Research Supervisor)

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Dr. Jorge Macedo, Ph.D. (Postdoctoral Research Supervisor)

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Dr. Krishna Kumar, Ph.D. (Ph.D. Supervisor)

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Dr. Ellen Rathje, Ph.D. (Ph.D. Committee Member)

Professor, Department of Civil, Architectural and Environmental Engineering, University of Texas at Austin, USA

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Dr. Jaehun Ahn, Ph.D. (M.S. Supervisor)

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